PIONEER JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2013

Candidate Name	

Civics Group						**INDEX NUMBER**					

H2 BIOLOGY **9648 / 03**

20 Sept 2013 **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Read the instructions on the answer sheet very carefully.
This document consists of 2 sections:

Section A
Answer **ALL** questions.
Answers are to be written in the spaces provided.

Section B
Answer Question 5.
Write your answer on the lines provided.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
1	/ 12
2	/ 15
3	/ 14
Section B	
5	/ 20
TOTAL	/ 60
4 (SPA)	/12

This document consists of **XX** printed pages (inclusive of this page)

[Turn Over

Section A: Answer all questions [40 marks]

1. The ability to identify salmon species following food processing such as canning and smoking is problematic but of great commercial importance. The most commonly used methods of identifying raw fish are not applicable to thermally treated products. Polymerase chain reaction-restriction fragment length polymorphism (PCR-RFLP) is thus an alternative method to authenticate canned or smoked fish products.

a) Suggest why authentication of processed fish product is important. [1]

- **to deter or identify mis-labelling so that consumers can make informed choices about their food.**
- **any other valid reasons**

Following extraction of DNA from the processed fish product, PCR was carried out to amplify a 500bp segment from the DNA sample. Figure 1.1 illustrates the thermal cycling characteristic of PCR.

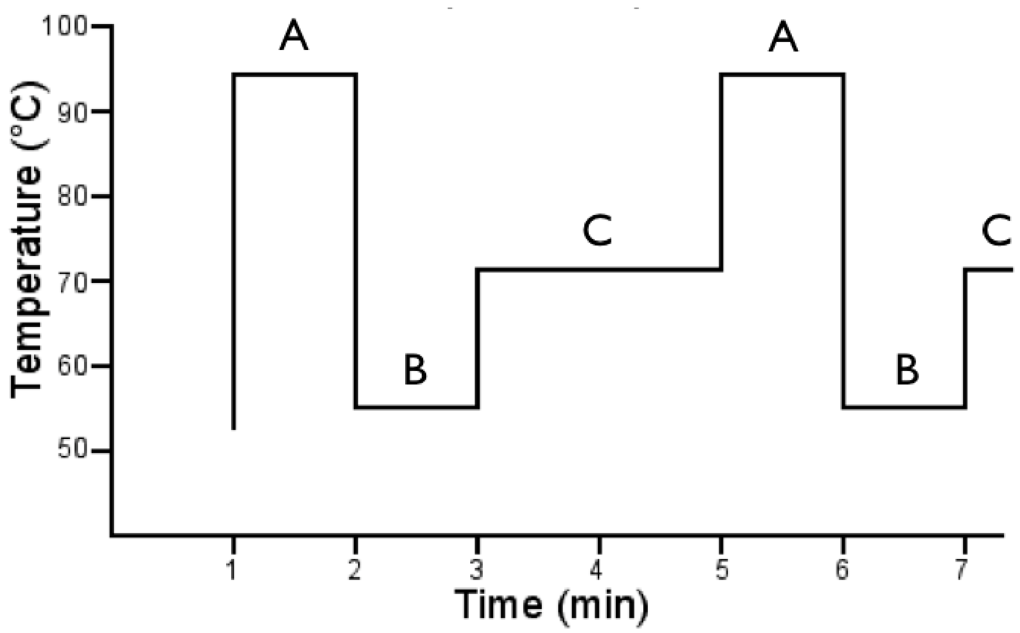


Figure 1.1

b) Explain the importance of the temperatures in phases A, B and C.

(i) Phase A [2]

- **At 95°C, hydrogen bonds between DNA strands are broken,**
- **DNA is denatured**
- **DNA double helix unwind and separated into two separate single strands**
- **Each used as a template for replication by DNA polymerase.**

(ii) Phase B [1]:

- **As specific DNA primers are used to anneal to 3' end of target sequence via complementary base-pairing,**
- **temperature is specific at 55°C;**

(iii) Phase C [1]

- **Optimal temperature for heat stable Taq polymerase is 72°C**
- ***Taq* polymerase will extend the primers by adding nucleotides to the 3' OH ends of the primers**

Figure 1.2 shows the number of molecules of DNA produced over a number of PCR cycles.

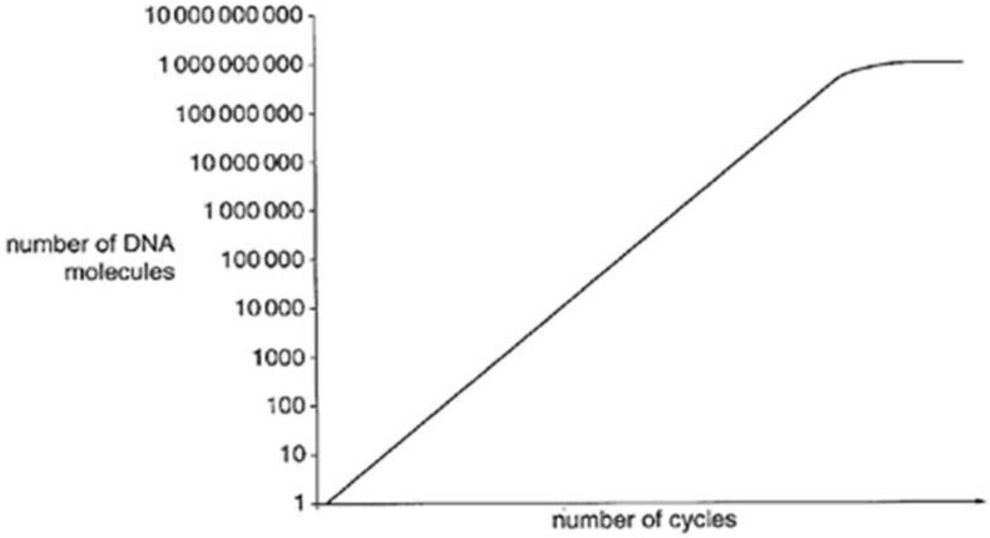


Figure 1.2

c) Suggest a reason why the graph plateaus. [1]

- **Taq polymerase gradually becomes denatured; OR**
- **No more primers or deoxyribonucleotides available;**

DNA samples were taken from 10 different smoked salmon products to authenticate the salmon used. The amplified samples were then subjected to restriction digests using various restriction enzymes. For each reaction, the DNA restriction fragments were subjected to gel electrophoresis and visualized using a staining kit.

Figure 1.3 shows the RFLP profiles of salmon products 1 to 10 following digestion with restriction enzymes Bsp 1286I, Eco RII and Sau 3AI. 100bp DNA ladder lanes were included as reference.

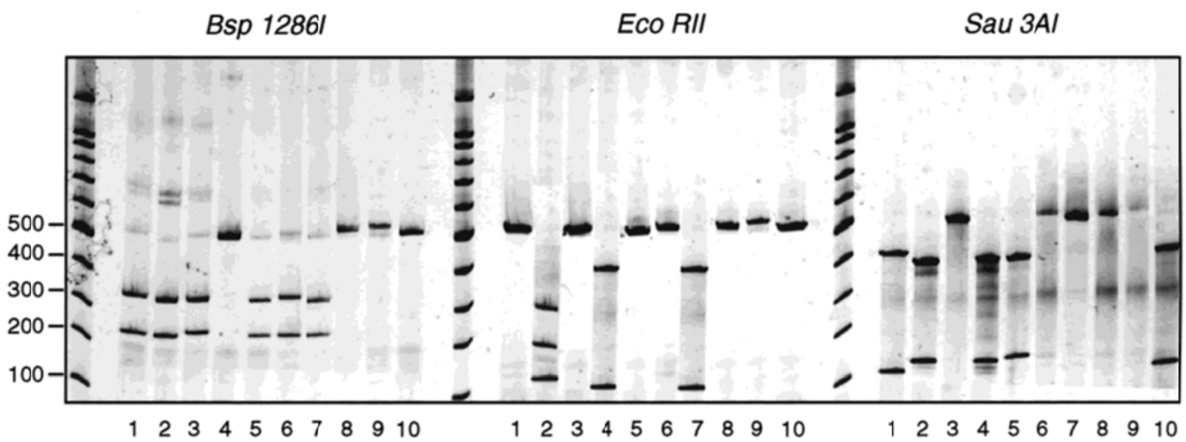


Figure 1.3

With reference to Figure 1.3,

d) explain what is meant by RFLP. [2]

- **Different length of fragments**
- **when genome of 2 individuals digested by the same RE**
- **Due to genetic mutation**
- **resulting in loss or gain of restriction sites**

e) identify and explain which 2 smoked salmon products use the same species of salmon. [4]

- **Salmon products 3 and 6;; [1 mark] or Salmon products 8 and 9**
- **For all three REs used, the RFLP profiles for both products are the same.**
- **Bsp 1286I -> both products show two bands of 200bp and 300bp. (500bp for 8 and 9)**
- **EcoRII -> one band of 500bp (500bp for 8 and 9)**
- **Sau3AI -> two bands of 300 and 500bp. (300bp and 500bo for 8 and 9)**
- **Genomes from both products are the same**
- **as the restriction sites for each respective RE are at the same loci.**

Question 2

(a) Explain the importance of ADA with respect to the immune system. [2]

- **ADA is important for breakdown of deoxyadenosine/ purine metabolism**
- **Prevent accumulation of the substrate deoxyadenosine in cells**
- **T lymphocytes will remain functional →**
- **Aids B lymphocytes in recognizing and producing antibodies**

(b) Suggest why a recessively inherited disorder such as ADA-SCID can be treated by gene therapy. [2]

- **Ref to single gene disorder;**
- **Loss-of-function mutation**
- **Introduction of one normal / dominant allele to produce sufficient amounts of gene product / ADA**

- **to mask effect of recessive mutation**

(c) (i) Suggest why the *ex vivo* method was used in this case. [1]

Either one:

- **T-lymphocytes can be easily transfected**
- **Larger amount of T-lymphocytes can be cultured in the laboratory**
- **Allow for selection of successfully transformed cells**
- **Prevent the retrovirus from targeting other rapidly proliferating cells**

Reject: viral vector can regain ability to cause disease

(ii) Suggest an improvement to the procedure in Figure 2.1 so as to increase the success rate of the treatment. [3]

- **culture blood stem cells / hematopoietic stem cells instead of T-lymphocytes**
- **ref. to origin of stem cells – bone marrow / umbilical cord**
- **differentiate to form all types of white blood cells (including B- and T-lymphocytes)**
- **Stem cells are able to self renew by mitotic cell division for a long period of time**
- **thus may provide more long term / permanent cure**

also accept:

- **select for / identify successfully transformed cells**
- **after infection of cells with retrovirus**
- **Increase transformed cells through cell culture before re-implant cells into patient**
- **Ensure that re-introduced cells are all able to express normal ADA gene**

(d) Name another method, other than gene therapy, which can be used for the treatment of ADA-deficiency. [1]

- **bone marrow transplantation from a perfectly matched sibling donor <u>OR</u>**
- **enzyme replacement therapy / weekly intramuscular injections of ADA**

(e) Gene therapy using the same form of treatment as the above case study has succeeded for 15 French baby boys suffering from X-linked SCID. However 3 of the little boys later developed cancer and died. Suggest how this is possible. [3]

- **inability to control the specific site of insertion / random integration of transgene -> insertional mutagenesis**
- **disrupt gene control regions -> activation of oncogenes**
- **Inactivation of 2 copies of the tumor suppressor gene**
- **Ref function of the TSG**
- **Inactivation of a gene involved in apoptosis**
- **-> uncontrolled cell division/proliferation**

Reject: statements that imply insertional inactivation of proto-oncogenes contributes to cancer.

(f) State and explain 2 other factors that hinder the effectiveness of current gene therapy approaches. [2]

1. **Short-lived nature of gene therapy**
   a. Problems with integrating therapeutic DNA into the genome and the rapidly dividing nature of many cells prevent gene therapy from achieving any long-term benefits.
   b. -> multiple rounds of gene therapy required
   c. Difficulty in inserting new genes into billions of target cells.
2. **Immune response to foreign object**
   a. Immune system's enhanced response to invaders it has seen before makes it difficult for repeated gene therapy.
3. **Problems with viral vectors**
   a. Viral vector may recover its ability to cause disease and trigger an immune response.
4. **Multigene disorders**
   a. Some commonly occurring disorders, such as heart disease, high blood pressure, Alzheimer's disease, arthritis, and diabetes, are caused by the combined effects of variations in many genes.
   b. Multigene or multifactorial disorders such as these would be especially difficult to treat effectively using gene therapy as multiple replacements of defective alleles required.
5. **Inability to control gene expression**
   a. Corrective genes may be over-expressed or a gene may shut off shortly after it has been introduced.
6. **Procedures for introducing DNA into cells typically are inefficient**
   a. Perhaps only 1 in 1,000 or 100,000 cells may receive gene of interest, thus a large population of cells is needed to attempt gene therapy.
7. **Fate of foreign DNA**
   a. In some cases the mutant gene is replaced by the normal gene, and in others the normal gene integrates randomly into the genome elsewhere. -> may result in insertional mutagenesis
   b. Ref TSG and proto-oncogenes and how mutation in these genes may lead to cancer development
8. **Adverse effects not known** if DNA is introduced into non-target cells and is expressed in these cells.
9. **Germline may be accidentally targeted** resulting in the inheritance of any adverse conditions.

3. AquAdvantage salmon is a transgenic salmon where a promoter for the anti-freeze gene of an ocean pout and a gene for the Chinook growth hormone was inserted into a plasmid pUC18 (Figure 3.1). The recombinant DNA (Figure 3.2) is then introduced into fertilized egg cells of the Atlantic salmon by microinjection and stimulated to become triploid.

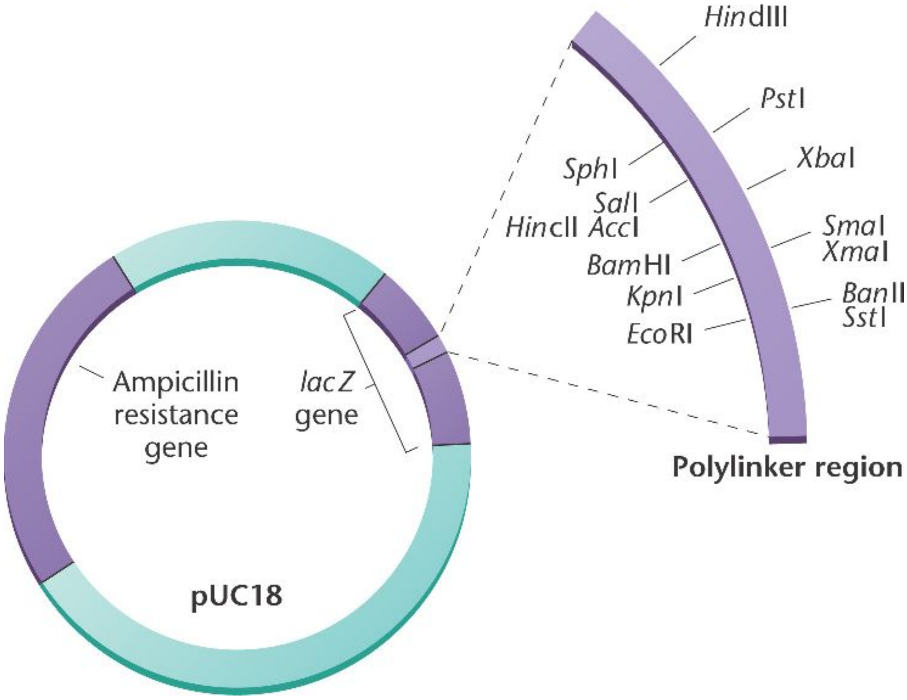


Figure 3.1

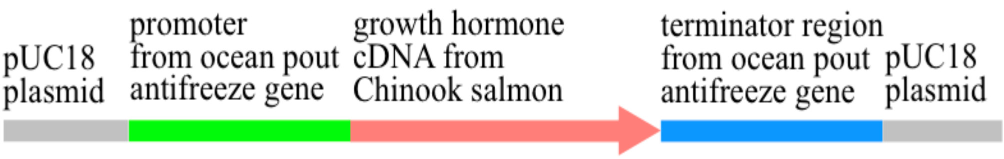


Figure 3.2

(a) With reference to Figures 3.1 and 3.2,

(a) describe the role of the ampicillin resistance gene. [1]

- **Acts as a genetic marker/selection marker**
- **to select for cells that have been transformed/taken up the pUC18 plasmid.**

(b) describe how the recombinant DNA is formed. [4]

- **mRNA from the growth hormone is converted to a single stranded cDNA using reverse transcriptase;**
- **the RNA from the RNA-cDNA hybrid is hydrolysed by RNAseH;**
- **DNA polymerase is used to form a complementary strand to form double stranded cDNA;**
- **Promoter and terminator region from ocean pout antifreeze gene is joined to the growth hormone cDNA using DNA ligase;**
- **Linker DNA is added to ends of the promoter and the terminator region;**

- **Linker DNA and pUC18 plasmid is cut using the same RE (state any RE shown in the diagram) to produce complementary sticky ends;**
- **Hydrogen bonds are formed/Annealing between the sticky ends;**
- **DNA ligase is used to form phosphodiester bonds between the transgene/gene construct to form a recombinant DNA.**

(b) The egg cells were induced to be triploid. Explain the significance of this. [2]

- **AquAdvantage salmon are infertile as unable to produce haploid eggs via meiosis;**
- **Any AquAdvantage salmon which might escape into the wild cannot mate with wild salmon**
- **hence preventing transfer of the transgene to wild species**

(c) Figure 3.3 below shows the differences between weight of AquAdvantage salmon and standard salmon.

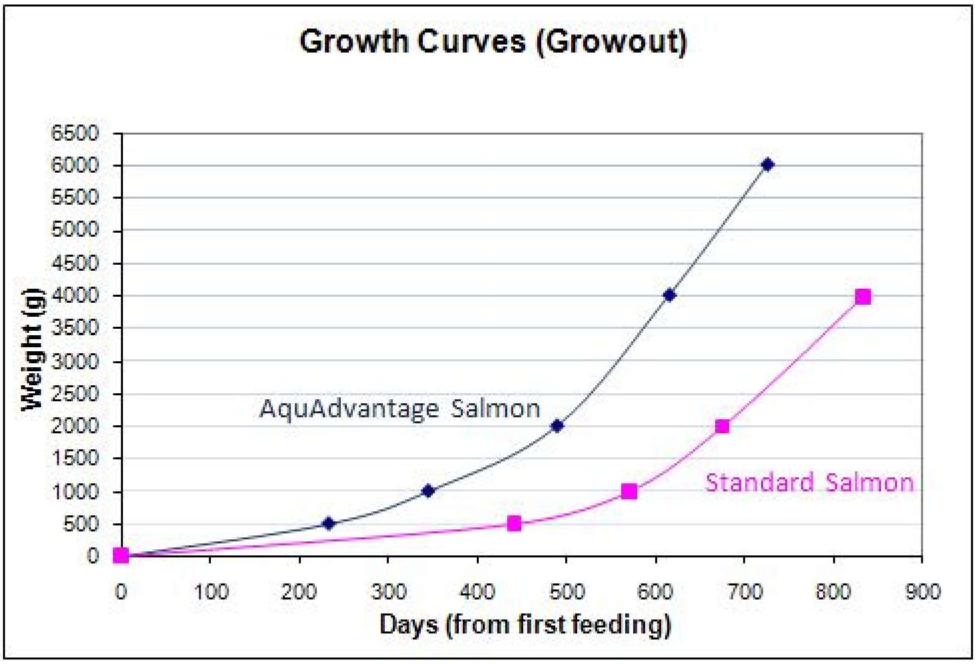


Figure 3.3

(i) With reference to Figure 3.3, compare the growth curves of standard salmon and AquAdvantage salmon. [3]

- **Both graphs show gradual increase and then exponential increase (necessary)**
- **AquAdvantage salmon is always heavier than standard salmon from the first feeding.**
- **Standard salmon reached maximum weight of 4000g in 850 days while AquAdvantage salmon reached more quickly 4000g in 620 days.**
- **The weight gain for AquAdvantage salmon was gradual from 0 to 600g from 0 to 230 days while standard salmon weight increased gradually from 0 to 600g from 0 to 440 days.**
- **AquAdvantage salmon showed an exponential increase in weight from 600g at 230 days to a maximum of 6000 g at 720 days. But standard salmon showed an**

exponential increase in weight from 600g at 440 days to a maximum of 4000g at 850 days.

or any other valid comparisons

(ii) Explain the difference in the weight increase of AquAdvantage salmon and the standard salmon. [2]

- **AquAdvantage salmon contains the growth hormone gene of the larger Chinook salmon -> produce the GH to grow larger;**
- **Promoter of the ocean pout antifreeze gene -> produce the hormone all year round;**
- **Enabling the AquAdvantage salmon to grow larger in a shorter period of time.**
- **Total weight gain from AquAdvantage salmon was 6000g over 720 days and standard salmon was 4000g in 850 days.**

(c) Describe one ethical and one social concern that might arise from the production of AquAdvantage salmon. [2]

Health concern (R: Environmental concern as the fish are triploid, cannot mate with wild type)

- **The effects of extra salmon growth hormones and the safety of these salmon for human consumption**

Ethical concern

- **Increased production of growth hormone has harmful effects on the health of salmon -> additional stress in salmon -> exploitation of animals for food**

or any other valid ethical and social concerns

4. **Planning Question [12 marks]**

The Tasmanian Tiger or Thylacine (*Thylacinus cynocephalus*), the world's largest carnivorous marsupial, was once common throughout Australia and Papua New Guinea. The Thylacine resembled a large, short-haired dog with a stiff tail which smoothly extended from the body in a way similar to that of a kangaroo. The female Thylacine had a pouch with four teats, but unlike many other marsupials, the pouch opened to the rear of its body.

An example of convergent evolution, the Thylacine showed many similarities to the members of the Canidae (dog) family of the Northern Hemisphere: sharp teeth, powerful jaws, raised heels and the same general body form.

Due to human activities, the Thylacine was hunted to extinction by early 1930. A Thylacine specimen with soft tissue remaining is found in the Australian Museum in Sydney.

Imagine that you are a researcher in the Australian Rare Fauna Research Association. Recently it was reported that locals near Mount Carstensz in Western New Guinea had sighted creatures that resemble Thylacines. Some members of the Association believe that the creatures sighted may be descendents of the Thylacine, while other members believe that the creatures may be a new species. If the former were true, then the Thylacine is not extinct and conservation effects may revive the species.

Plan an investigation to investigate if these creatures found in Western New Guinea are descendents of the Thylacine that was thought to be extinct.

Your planning must be based on the assumption that you have been provided with the following equipment and materials.

- Tissue sample from the museum specimen and from the Western New Guinea creatures under investigation
- Pestle and mortar
- DNA extraction buffer solution
- Microcentrifuge tubes
- Centrifuge
- Restriction enzymes
- Agarose or polyacrylamide gel plate
- Suitable source of electric current
- Radioactive probe
- Nitrocellullose membrane
- Autoradiography equipment

Your plan should have a clear and helpful structure to include:

- An explanation of the theory to support your practical procedure
- A description of the method used including scientific reasoning behind the method
- The type of data generated by the experiment
- How the results will be analysed including how the origin of the organism can be determined

Suggested answer

Introduction: 4 marks

- Use of molecular homology: the similarities in DNA nucleotides sequence of organisms
- Especially certain VNTR / DNA sequences that are conserved within a species but different between species.
- Extract the DNA from the specimen from the nucleus
- Brief description of RFLP and reference to DNA fingerprinting
- Use of a suitable RE to cut the genomic DNA – recognise specific RE sites within the DNA (to ensure that RE used does not cut up the target DNA sequence
- Separate fragments using gel electrophoresis via molecular sizes
- Perform southern blotting (brief outline of blotting process and denaturation of double stranded DNA);;
- Nucleic Acid hybridisation using a single stranded radioactive DNA probe (DNA if VNTR is used) that is complementary to the target sequences
- Description of autoradiography – use of xray film

Hypothesis:

- If the RFLP banding patterns obtained are identical or similar, it suggests that the creatures found in Western New Guinea are closely related to the Thylacine, thus this suggests that they may share a recent common ancestor.

Procedures: (7 marks)
Proposed method of experiments to achieve results:

(DNA extraction → digestion with RE → gel electrophoresis → Southern blotting)

(i) Method of DNA extraction from tissue samples including use of buffers;;

- Mechanical breakage of tissues with pestle and mortar
- Use of DNA extraction buffer solution
- Centrifuge & DNA in the supernatant

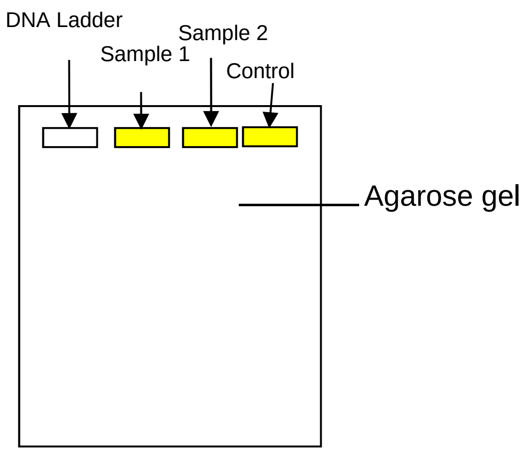
(ii) Digestion with a suitable restriction enzyme for 1 hour at optimum temperature of 37^0C. (Restriction enzyme must be able to cleave the DNA sequence at specific sites generating different sized fragments between species.);;

- 2µl DNA
- 0.5µl - 1 µl restriction enzyme

(iii) Preparation of samples for electrophoresis (3 marks)

- Preparation of agarose gel – dissolve 1g of agarose powder in 1L of electrophoresis buffer solution (1% gel)
- Add loading dye eg: bromophenol blue to DNA samples:
- First lane –20 µl DNA ladder
- Lane 2-4 - Sample 1: 20 µl Restriction digest of DNA from museum specimen
- Lane 5-7 - Sample 2: 20 µl Restriction digest of DNA from creatures under investigation)
- Lane 8- Prepare a control tube with all the reagents except DNA (Blank).

(iv) Load DNA samples in gel as shown:  - represents 3 lanes of same sample



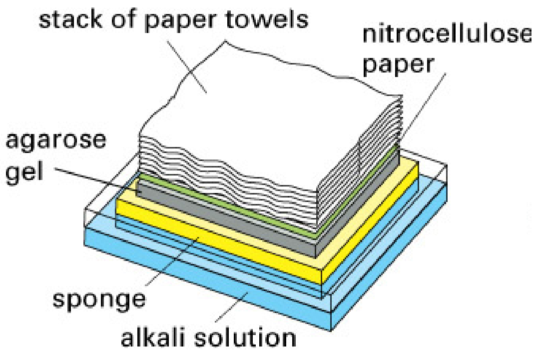
(v) Separation of fragments by gel. Run at 110V for 30min. Stop current when loading dye reach 2/3 gel;

(vi) Transfer of DNA on to nitrocellulose membrane (blotting process);

- Gel covered with nitrocellulose membrane, adsorbent papers added on top of membrane, single stranded DNA drawn up gel by capillary action

(vii) Hybridization with radioactive labelled DNA probe;

- Alkali solution to denature DNA fragments into single strands
- Radioactive probe binds by complementary base pairing to DNA sequence of interest
- Excess probes are washed off



SEPARATED DNA FRAGMENTS
BLOTTED ONTO NITROCELLULOSE PAPER

(viii) Autoradiography method;

- A photographic film or an X ray film is laid over the filter. Radioactivity in the bound probe exposes the film.
- DNA that has hybridised to the labeled probe shows up as bands on the autoradiograph and can be visualized

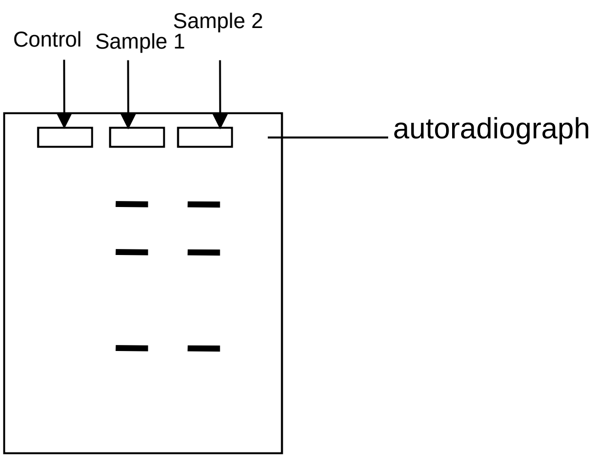
(ix) **Repeat the entire experiment twice** using fresh samples;

Control;

Tube with all reagents but no DNA added. This negative control will show that only DNA fragments with regions complementary to the probe will show up as a dark band after autoradiography.

.

More tests can be done on other VNTR / gene sequences to obtain more evidence of homology.



Safety precautions;

- be careful when working with radioactive DNA probes – wear goggles, gloves and lab coat to prevent touching bare skin

Section C: Essay question

Write your answers to this question on the lines provided.

Your answer:

- Should be illustrated by large, clearly labelled diagrams, where appropriate.
- Must be in continuous prose, where appropriate.
- Must be set out in section (a), (b), etc., as indicated in the question.

(a) Describe, using a named example, how the micropropagation technique is used in production and cloning of herbicide resistant plants. [9]

a) **Obtain mRNA for glyphosate-resistant EPSPS gene and using reverse transcription method, produce the double stranded cDNA *(or any sound method to isolate this gene)***
   - **Add linker molecules to both ends of the gene of interest**

b) **Remove the oncogenes and opine producing genes from the Ti plasmids of Agrobacterium.**

c) **cut the Ti plasmid using the same restriction enzymes as that for Bt gene, forming complementary sticky ends**

d) **Mix the cut plasmid and gene of interest to allow complementary base-pairing via hydrogen bonds to temporarily hold the fragments together**

e) **Add DNA ligase to seal the nicks in the sugar-phosphate backbone forms phosphodiester linkages between fragments**

f) **Insert recombinant Ti-plasmids back into the Agrobacterium**

g) **In micropropagation, plant tissues or explants eg: meristematic cells are removed from plants ® explants only;**

h) **Infect the explants with with Agrobacterium containing the recombinant Ti plasmids. Select explants that were successfully transformed;**

i) **Surface of explants sterilised with dilute sodium hypochlorite (Clorox) to kill bacterial and fungal pathogens or organisms;**

j) **And grown in a containing sterile media containing nutrients and plant growth regulators needed for plant growth;**

k) **Containers holding the explants are then sealed and incubated for 1-9 weeks.**

l) **During this period, the cultured cells divide by mitosis**

m) **to form a mass of undifferentiated tissue called a callus.**

n) **As callus increases in size, pieces of callus is sliced off and grown on new medium composition (subculturing);**

o) **By adjusting concentration of auxin:cytokinin ratio in growth medium,**

p) **cells in callus can be induced to differentiate into roots and shoots;**

q) **Further growth being encouraged by the use of plant growth substances**

until plantlets are large enough

r) **To be weaned and planted in sterile soil in green house**

s) **After acclimatization in green house, the plant are transferred to soil for field planting**

b) Discuss the social and ethical implications of GMO plants using 3 named examples. [6]

1) Definition of GMO – 1m
Genetically Modified Organism: Organism that has acquired one or more genes by artificial means. The genes may or may not be from the same species.

Three named examples + description (each 1 mark)

2) Social implications
(a) Threat to Human Safety
(i) Genetic markers

- Vectors used for transforming plant/crop cells – contain antibiotic markers/ kanamycin resistance;
- Concerns that the antibiotic resistant gene if it is not properly broken down by the digestive system may be passed to potentially harmful bacteria in the human gut making them resistant to the antibiotics;

(b) Threat to Safety of the Environment
(i) Genetically modified crops might establish themselves as weeds (Bt corn, Roundup Ready soybean, any valid e.g.)

- Seeds from the transgenic crops that have been modified to withstand unfavourable environmental conditions, may be carried by the wind to other places. These crops grow quickly and may establish themselves as weeds;

(ii) Spread of resistance from genetically modified crops to weeds (Any herbicide resistant plant)

- Pollen grains from genetically modified plants can be carried in the wind and fertilise wild relatives and weeds;
- -> "superweeds". These may become more invasive (overrun agricultural areas rapidly) or compete with the natural species;

(iii) Upsetting the ecological balance (GM salmon, Bt corn etc.)

- the genetically modified organisms themselves may be more aggressive/ out-compete other plants in the same area;
- -> may upset the balance of the native species / decrease in biodiversity;

3) Ethical implications
(a) Genetic manipulation of plants may be deemed unacceptable as there are no rights to tampering with nature and affecting the intrinsic value of the plant ;

(b) Is it acceptable to engineer any organism to be used as food for human consumption. (Give e.g.)

- Plants are not biologically 'designed' to withstand the stress of producing additional proteins / toxins;

Golden rice/Bt Corn

(c) Genetic engineering to make a staple crop more resistant in marginal conditions could be a danger of increased dependence on rich "developed" countries

- i.e. richer countries benefiting at the expense of the poorer countries;
- world food production may be dominated by a small number of large companies with the technical know-how;

Other points:

(d) Non-mandatory labeling of products containing GMO plants -> consumer rights;

(e) Patent rights to GMO plants -> patenting plants is itself unethical as it reduces them to the level of objects;

(c) Discuss the limitations for the use of gene therapy in treating diseases in humans. [5]

A) Difficulties in ensuring appropriate regulation / activity / uptake of gene

1. ref. cells make appropriate amounts of gene product / low levels of expression
2. worsen by low efficiency of uptake of DNA by target cells
3. inserted genes in heterochromatic region may not be expressed
4. Right time and in the right location of the body not ascertained
5. Technical limitations in producing large numbers of retroviral vectors
6. Non-target cells may take up gene

B) Immune response against the introduced transgenic cell or vector

1. Repeated therapies may thus be ineffective
2. Danger of viral vector recovering its ability to cause disease in the patient
3. ref. difficulty controlling if the transgene is correctly integrated into the genome -> insertional mutagenesis
4. development of cancer -> ref. tumour suppressor gene or proto-oncogene

C) Difficulty in treating multigene disorders

1. require multiple replacement of defective alleles
2. ref. one named disorder: cancer, heart disease, high blood pressure, Alzheimer's disease, arthritis, diabetes etc

D) Short-lived nature of gene therapy

1. target cells may be short-lived or may become unstable after introduction of therapeutic DNA
2. many rounds of gene therapy required for a long period of time
3. rapidly dividing nature of cells prevents efficiency of treatment if gene is extrachromosomal